EXPERIMENT-05

K-Nearest Neighbours (K-NN): Dataset-Based Python Programs

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5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Problem 1: Flower Classification

Aim

To implement the K-Nearest Neighbours algorithm on a flower dataset stored in a CSV file and classify a new flower sample using k = 3.

Procedure

1. Save the training examples in flower_data.csv with the class label in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV file and choose the value of k and the new sample to be classified.
4. Calculate the Euclidean distance between the new sample and every training sample.
5. Select the k training samples with the smallest distances.
6. Assign the class that occurs most often among the k neighbours and print it.

Dataset File / Input

File name: flower_data.csv

The CSV file contains a header row followed by these training examples:

PetalLength	PetalWidth	Species
1.4	0.2	Setosa
1.3	0.2	Setosa
1.5	0.2	Setosa
1.7	0.4	Setosa
1.4	0.3	Setosa
4.7	1.4	Versicolor
4.5	1.5	Versicolor
4.9	1.5	Versicolor
4.0	1.3	Versicolor
4.6	1.5	Versicolor
6.0	2.5	Virginica
5.1	1.9	Virginica
5.9	2.1	Virginica
5.6	1.8	Virginica
6.1	2.3	Virginica

Code

```
import numpy as np
import pandas as pd
from collections import Counter

df = pd.read_csv("flower_data.csv")
X = df.iloc[:, :-1].values.astype(float)
y = df.iloc[:, -1].values

new = np.array([5.7, 2.0])                              # new sample to classify
```

```
k = 3

dist = np.sqrt(((X - new) ** 2).sum(axis=1))       # Euclidean distance
nearest = dist.argsort()[:k]                       # indices of k nearest neighbours

print("New sample:", new)
for i in nearest:
    print("Neighbour:", X[i], "Class:", y[i], "Distance:", round(dist[i], 2))
print("Predicted class:", Counter(y[nearest]).most_common(1)[0][0])
```

Output

```
New sample: [5.7 2. ]
Neighbour: [5.9 2.1] Class: Virginica Distance: 0.22
Neighbour: [5.6 1.8] Class: Virginica Distance: 0.22
Neighbour: [6.1 2.3] Class: Virginica Distance: 0.5
Predicted class: Virginica
```

Result

The new sample (PetalLength = 5.7, PetalWidth = 2.0) has its three nearest neighbours at distances 0.22, 0.22 and 0.5, and all three are Virginica. The predicted class is Virginica.

Problem 2: Student Result

Aim

To classify a new student as Pass or Fail from Maths and Science marks using K-NN (k = 3) on a dataset stored in a CSV file.

Procedure

1. Save the training examples in student_data.csv with the class label in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV file and choose the value of k and the new sample to be classified.
4. Calculate the Euclidean distance between the new sample and every training sample.
5. Select the k training samples with the smallest distances.
6. Assign the class that occurs most often among the k neighbours and print it.

Dataset File / Input

File name: student_data.csv

The CSV file contains a header row followed by these training examples:

MathsMarks	ScienceMarks	Result
85	80	Pass
78	82	Pass
90	88	Pass
70	75	Pass
65	72	Pass
80	68	Pass
35	40	Fail
42	38	Fail
30	45	Fail
48	35	Fail
38	50	Fail
45	42	Fail

Code

```
import numpy as np
import pandas as pd
from collections import Counter
```

```
df = pd.read_csv("student_data.csv")
X = df.iloc[:, :-1].values.astype(float)
y = df.iloc[:, -1].values

new = np.array([50, 45])                              # new sample to classify
k = 3

dist = np.sqrt(((X - new) ** 2).sum(axis=1))       # Euclidean distance
nearest = dist.argsort()[:k]                       # indices of k nearest neighbours

print("New sample:", new)
for i in nearest:
    print("Neighbour:", X[i], "Class:", y[i], "Distance:", round(dist[i], 2))
print("Predicted class:", Counter(y[nearest]).most_common(1)[0][0])
```

Output

```
New sample: [50 45]
Neighbour: [45. 42.] Class: Fail Distance: 5.83
Neighbour: [48. 35.] Class: Fail Distance: 10.2
Neighbour: [42. 38.] Class: Fail Distance: 10.63
Predicted class: Fail
```

Result

The new sample (Maths = 50, Science = 45) has three Fail neighbours at distances 5.83, 10.2 and 10.63. The predicted class is Fail.

Problem 3: Customer Segment

Aim

To assign a new customer to the Premium or Budget segment from age and spending score using K-NN (k = 3) on a dataset stored in a CSV file.

Procedure

1. Save the training examples in customer_data.csv with the class label in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV file and choose the value of k and the new sample to be classified.
4. Calculate the Euclidean distance between the new sample and every training sample.
5. Select the k training samples with the smallest distances.
6. Assign the class that occurs most often among the k neighbours and print it.

Dataset File / Input

File name: customer_data.csv

The CSV file contains a header row followed by these training examples:

Age	SpendingScore	Segment
22	85	Premium
25	78	Premium
30	90	Premium
28	72	Premium
35	80	Premium
24	92	Premium
45	20	Budget
52	15	Budget
48	30	Budget
58	10	Budget
40	25	Budget
50	35	Budget

Code

```
import numpy as np
import pandas as pd
from collections import Counter

df = pd.read_csv("customer_data.csv")
X = df.iloc[:, :-1].values.astype(float)
y = df.iloc[:, -1].values

new = np.array([32, 60])                                  # new sample to classify
k = 3

dist = np.sqrt(((X - new) ** 2).sum(axis=1))       # Euclidean distance
nearest = dist.argsort()[:k]                       # indices of k nearest neighbours

print("New sample:", new)
for i in nearest:
    print("Neighbour:", X[i], "Class:", y[i], "Distance:", round(dist[i], 2))
print("Predicted class:", Counter(y[nearest]).most_common(1)[0][0])
```

Output

```
New sample: [32 60]
Neighbour: [28. 72.] Class: Premium Distance: 12.65
Neighbour: [25. 78.] Class: Premium Distance: 19.31
Neighbour: [35. 80.] Class: Premium Distance: 20.22
Predicted class: Premium
```

Result

The new sample (Age = 32, SpendingScore = 60) has three Premium neighbours at distances 12.65, 19.31 and 20.22. The predicted segment is Premium.

Problem 4: Rain Prediction

Aim

To predict whether it will rain from temperature and humidity using K-NN (k = 3) on a weather dataset stored in a CSV file.

Procedure

1. Save the training examples in weather_data.csv with the class label in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV file and choose the value of k and the new sample to be classified.
4. Calculate the Euclidean distance between the new sample and every training sample.
5. Select the k training samples with the smallest distances.
6. Assign the class that occurs most often among the k neighbours and print it.

Dataset File / Input

File name: weather_data.csv

The CSV file contains a header row followed by these training examples:

Temperature	Humidity	Rain
24	90	Yes
22	85	Yes
26	92	Yes
21	88	Yes
25	80	Yes
23	95	Yes
34	40	No
36	35	No
32	45	No
38	30	No
33	50	No
35	42	No

Code

```
import numpy as np
import pandas as pd
from collections import Counter

df = pd.read_csv("weather_data.csv")
X = df.iloc[:, :-1].values.astype(float)
y = df.iloc[:, -1].values

new = np.array([30, 60])                                # new sample to classify
k = 3

dist = np.sqrt(((X - new) ** 2).sum(axis=1))       # Euclidean distance
nearest = dist.argsort()[:k]                       # indices of k nearest neighbours

print("New sample:", new)
for i in nearest:
    print("Neighbour:", X[i], "Class:", y[i], "Distance:", round(dist[i], 2))
print("Predicted class:", Counter(y[nearest]).most_common(1)[0][0])
```

Output

```
New sample: [30 60]
Neighbour: [33. 50.] Class: No Distance: 10.44
Neighbour: [32. 45.] Class: No Distance: 15.13
Neighbour: [35. 42.] Class: No Distance: 18.68
Predicted class: No
```

Result

The new sample (Temperature = 30, Humidity = 60) has three neighbours that all have Rain = No, at distances 10.44, 15.13 and 18.68. The predicted class is No.

Problem 5: Fruit Type

Aim

To classify a new food item as Fruit, Vegetable or Protein from its sweetness and crunchiness using K-NN (k = 3) on a dataset stored in a CSV file.

Procedure

1. Save the training examples in fruit_data.csv with the class label in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV file and choose the value of k and the new sample to be classified.
4. Calculate the Euclidean distance between the new sample and every training sample.
5. Select the k training samples with the smallest distances.
6. Assign the class that occurs most often among the k neighbours and print it.

Dataset File / Input

File name: fruit_data.csv

The CSV file contains a header row followed by these training examples:

Sweetness	Crunchiness	Type
10	9	Fruit
10	1	Fruit
7	3	Fruit
8	5	Fruit
9	2	Fruit
7	10	Vegetable
3	10	Vegetable
3	7	Vegetable
2	8	Vegetable
4	9	Vegetable
1	4	Protein
1	1	Protein
3	6	Protein
2	3	Protein
1	2	Protein

Code

```
import numpy as np
import pandas as pd
from collections import Counter

df = pd.read_csv("fruit_data.csv")
X = df.iloc[:, :-1].values.astype(float)
y = df.iloc[:, -1].values

new = np.array([6, 4])                              # new sample to classify
k = 3

dist = np.sqrt(((X - new) ** 2).sum(axis=1))       # Euclidean distance
nearest = dist.argsort()[:k]                        # indices of k nearest neighbours

print("New sample:", new)
for i in nearest:
    print("Neighbour:", X[i], "Class:", y[i], "Distance:", round(dist[i], 2))
print("Predicted class:", Counter(y[nearest]).most_common(1)[0][0])
```

Output

```
New sample: [6 4]
Neighbour: [7. 3.] Class: Fruit Distance: 1.41
Neighbour: [8. 5.] Class: Fruit Distance: 2.24
Neighbour: [9. 2.] Class: Fruit Distance: 3.61
Predicted class: Fruit
```

Result

The new sample (Sweetness = 6, Crunchiness = 4) has three Fruit neighbours at distances 1.41, 2.24 and 3.61. The predicted class is Fruit.